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Exploring modifiable neighborhood risk factors for fatal opioid overdose: A case-control study in two US cities

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Abstract

To explore associations between physical and social neighborhood factors and fatal opioid overdose, we remotely visited 2018–2019 fatal opioid overdose locations in New York City (n=2867) and Chicago (n=1677) via Google Street View and used a reliable and valid tool to assess 65 street block characteristics. We compared these locations to a proportional sample of blocks with no 2018–2019 overdoses (New York City n=2093; Chicago n=1148). We used logistic regression to explore associations between block characteristics and odds of an overdose event, controlling for neighborhood-level covariates (poverty, segregation). For both cities, blocks had significantly increased odds ($p<0.05$) of being overdose case sites if they had apartment buildings, bus stops, street trash, traffic calming features, and warning signs. New York City blocks also had significantly increased overdose odds if they had multifamily homes, commercial businesses, poor sidewalk maintenance, and loitering, and significantly decreased odds if they had single family homes, row homes, and security alarm signs. Chicago blocks with significantly increased overdose odds had vacant lots, abandoned buildings, alleys, restaurants, and adults on the street and significantly decreased odds with landscaping. Findings support neighborhood social and physical characteristics as important risk factors for fatal opioid overdose over and above sociodemographics.

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Keywords

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INTRODUCTION

The rate of U.S. drug overdose deaths nearly quadrupled from 2002 to 2023,¹ and 78% of drug overdose deaths in 2023 involved opioids.^{2,3} The opioid overdose rate in urban counties currently surpasses rural counties, and urban areas report higher overdose rates involving heroin, synthetic opioids (fentanyl), and cocaine.⁴ At the same time, there are significant disparities in fatal overdose rates across urban neighborhoods; for example, in 2022, 14 neighborhoods in New York City had overdose mortality rates higher than the city average, and five neighborhoods in the South Bronx had rates more than double the city average.⁵ Previous studies of overdose in urban areas point to neighborhood social and physical characteristics as important risk factors for fatal opioid overdose.^{6–10} Further inquiry into neighborhood context is critical to reduce fatal opioid overdose disparities across urban neighborhoods.

A growing body of evidence suggests that characteristics of the neighborhood environment significantly impact drug use, including injection drugs,¹¹ crack-cocaine,^{12,13} and marijuana,^{14,15} as well as fatal and nonfatal opioid overdose.^{6–8,16} These characteristics can be subdivided into macro and micro characteristics. Macro characteristics include aspects of overall community design related to land use, zoning, residential density, and other policies that can be difficult to change quickly.^{17(p1275)} Macro elements also include broader community social processes such as social capital and collective efficacy—resources stemming from the structure of neighborhood social relationships, which in turn facilitate the achievement of community goals.^{18,19} Higher neighborhood social capital is inversely associated with county-level opioid overdose mortality.²⁰ Micro characteristics are infrastructure that is smaller in scale and generally changeable more rapidly and with less cost.^{17(p1275)} Micro features include physical disorder—the deterioration of the urban landscape (e.g., graffiti, litter)—and social disorder—behavior which may be considered threatening (e.g., verbal street harassment, public intoxication).²¹ Previous studies have found significant relationships between specific neighborhood microfeatures and population-level opioid overdose rates, including abandoned or dilapidated housing^{6,8} and liquor stores.²² To our knowledge, this is the first study to examine neighborhood micro aspects related to opioid overdose by comparing blocks where fatal opioid overdoses occurred to blocks with no fatal opioid overdoses in the same time period.

The street block is a key mediating structure linking larger neighborhood ecological processes with responses of individuals and small groups of residents.²³ In other words, the broader social, political, and environmental characteristics of communities are measurable via the physical and social characteristics of street blocks.²³ Systematic social observation (SSO) is a standardized approach for directly observing neighborhood physical, social, and economic characteristics.²¹ Observations of naturally-occurring events or characteristics are recorded according to specific rules to facilitate replication; the SSO method is

particularly useful for assessing neighborhood physical conditions and social interactions that may not be accurately captured or described in surveys or interviews.²¹ Previous research into neighborhood microfeatures and population-level opioid overdose rates used surveys with self-reported measures of perceived neighborhood disorder.^{6,7,11,13,24,25} These measures may suffer from “same-source bias,” meaning that respondents may answer differently based on their own behavior.²⁶ Neighborhood demographics may also influence bias in self-reported measures; for example, people may be more inclined to evaluate neighborhoods with higher proportions of low-income or Black residents as having high levels of disorder regardless of actual levels of disorder because of historical and deeply entrenched stigmatization.²⁷

The purpose of this study was to characterize the associations between physical and social neighborhood factors and fatal opioid overdose. In this exploratory study, we operationalized neighborhood micro characteristics to identify visible features of the urban street block which may influence where and how people use drugs, controlling for macro-level processes such as poverty and segregation. Understanding neighborhood features associated with fatal opioid overdose may identify actionable community development strategies to reduce opioid use and prevent overdose, such as vacant lot or abandoned building remediation.^{28–30} We hypothesized that we would identify discernable street block-level patterns unique to locations where fatal opioid overdose occurred compared to blocks where no overdoses occurred over the same time period based on previous studies of neighborhood-level overdose risk factors (see Figure S1 for conceptual model).^{6–8,10,22}

METHODS

Study design and site description

This unmatched case-control study compared street blocks where opioid overdoses occurred from 2018–2019 (cases) to street blocks where no opioid overdoses occurred during the same period (controls). The unit of analysis was the street block, outcome was case vs. control status, and exposures were social and physical environmental conditions of the street blocks.

The two metropolitan areas included in this study—New York City and Chicago—have populations over 2 million and steadily increasing fatal overdose rates since 2010.^{2,3,31,32} These cities represent diverse geographic regions (Northeast, Midwest) with disparate urban landscapes, population densities, and economic climates. Investigating neighborhood context for overdose prevention across these cities will potentially increase generalizability of findings.

Data sources

Cases: Fatal opioid overdose data, 2018–2019: New York City Office of the Chief Medical Examiner (OCME) provided records on all accidental drug overdose deaths occurring within the five boroughs (Manhattan, Bronx, Brooklyn, Queens, Staten Island). Medical examiners are appointed and have board-certification in a medical specialty.³³ During their investigation into cause of death, medical examiners review a variety of data

including autopsy reports, toxicology, interviews with friends and family, and police, EMS, and hospital reports; they may also visit the overdose location.³⁴ OCME records included toxicology reports, full street address where the overdose occurred, and demographics; records determined by OCME to be suicides or homicides were not provided. We identified opioid-involved overdoses using text-based identification of drug involvement³⁵—illegal or prescription opioids (e.g., heroin, oxycodone) and fentanyl or fentanyl metabolites (n=2,867). Addresses of overdose locations were geocoded to latitude and longitude points from street addresses and mapped in ArcGIS 10.4.

Cook County Medical Examiner (ME) records are publicly available and updated daily³⁶ and include full toxicology reports and GPS coordinates for the exact location where the overdose occurred. We excluded overdoses outside the Chicago city limits to maintain comparable urbanicity between New York City and Chicago. We identified opioid-involved fatal overdoses using text-based identification (n=1,677).³⁵ GPS coordinates of overdose locations were mapped in ArcGIS 10.4. Records labeled suicides or homicides by the ME were excluded from the analysis.

Demographic information for both cities included age, sex, and race/ethnicity. We recoded race/ethnicity into four categories: non-Latino white, non-Latino Black, Latino, and Other (combining Asian/Pacific Islander, “other,” and “unknown”). We excluded overdoses that did not involve any opioids because environmental context and individual characteristics may differentially impact drug choice, route of administration, and associated overdose risk.^{37–39}

Control block selection: We used publicly available planimetric maps of New York City and Chicago to generate a street block sampling frame. A street block is defined as the distance from one intersection to the next intersection, a distance of approximately 0.1 miles (160 m).⁴⁰ We selected control blocks to be proportionally similar to overdose distribution by New York City borough or Chicago region (Downtown: Chicago Central and Loop, Chicago South Side, Chicago West Side, Chicago North Side). For example, 24.5% of New York City overdoses occurred in Manhattan, so a similar proportion of controls blocks were selected from Manhattan. This ensured that blocks were selected across the cities as opposed to concentrated in areas with lower overdose frequencies (Figure S2). We calculated 2,000m buffers around each case location in Python 3.10—the upper bound of walking distance for most nonrecreational urban walking trips^{41,42}—to create a sampling frame of potential control blocks similarly distributed across the cities compared to cases. We then used a random number generator to select control blocks. We excluded blocks categorized as expressways or ramps, as well as pedestrian-only roadways or walking trails through parks as Google Street View (GSV) images were not available for these block types. Control blocks $\leq 75\text{m}$ were excluded as were consecutive blocks (e.g., if a case was on the 200 block of Main Street, the control could not be the 100 or 300 blocks of Main Street). This allowed for discrete blocks without overlap between cases and controls. If members of the data collection team reported a block fell into one of the excluded categories, or if GSV images were not available, the block was replaced by the next control block in the randomly ordered list.

Built and social environment original data collection: Neighborhood Inventory for Environmental Typology (NifETy) is a SSO tool designed to assess neighborhood characteristics related to violence, alcohol, and other drug exposures (Table S1).⁴³ NifETy has been used in previous studies to examine the impact of neighborhood characteristics on drug use risk factor.^{14,44,45} NifETy was originally designed for in-person data collection and included over 160 items operationalized into seven domains (e.g., physical and social disorder).⁴³ NifETy shows strong reliability and validity metrics.⁴⁶ NifETy was adapted for use in GSV to allow for faster and more cost-efficient data collection.⁴⁷ GSV is a free tool offering panoramic, street-level images; the user types in an address and can virtually “walk” forward or backward along a street, revolve 360 degrees, rotate vertically 290 degrees, and zoom in and out. GSV images are time-stamped with month and year an image was processed, and the user can travel back in time to every previous image. Inter-rater reliability metrics were strong for the majority of NifETy items in GSV (ICC≥0.7), and items were highly correlated with in-person observations ($r\geq 0.6$) in previous studies.⁴⁷

Full details of GSV data collection protocol are presented in Nesoff et al.⁴⁷ Briefly, undergraduate interns participated in a two-hour data collection protocol training, then practiced data collection on 10 non-study blocks. Project managers evaluated practice data for inaccuracies and met with interns for an additional one-hour review. During data collection, project managers randomly chose three data points from each intern weekly to monitor data quality and consistency across raters. Project managers held weekly meetings to answer intern questions and monitored a group discussion forum to provide real-time support.

For cases, NifETy items were collected on the block containing the address of each fatal overdose using images time stamped within six months of the overdose. Interns entered the exact GPS coordinates for the overdose location into GSV and virtually “walked” the block to collect NifETy items (Figure S3). For controls, we randomly assigned each block to one of four six-month intervals (January-June 2018; July-December 2018; January-June 2019; July-December 2019); interns used images time stamped within the assigned interval. If no image was available in the six-month interval, interns coded the image closest in date to the interval with a preference for images before the interval (e.g., if images were available eight months before or eight months after the interval, interns chose the images from eight months before). Interns collected NifETy items on the entire control block by virtually “walking” from intersection to intersection. Data collection for case blocks took place June-August 2021 and for control blocks in January 2022. The University of Pennsylvania institutional review board approved this study.

Measures

Demographics for New York City and Chicago census block groups were taken from five-year American Community Survey (ACS) estimates for each year of data (e.g., 2019 ACS estimates were paired with 2019 cases and controls).⁴⁸ We used ArcGIS’ spatial join tool to identify the census block group containing each case and control block and assigned the corresponding block group measures to the case or control. We calculated neighborhood deprivation with the formula $\{((c/10+d/10)-(a/10+b/10))/4\}$ using census

block group-level ACS items: (a) adults ≥ 25 years with a college degree, (b) owner-occupied housing, (c) households with incomes below the federal poverty threshold, and (d) female-headed households with children (percentages are entered as whole numbers, not decimals); range=[-5 is very low/little deprivation, +5 is very severe deprivation].⁴⁹ We assessed segregation using Index of Concentration at the Extremes (ICE) by subtracting the number of non-Latino Blacks from the number of non-Latino whites in a block group and dividing by block group population; range=[-1 is 100% Black; 0 is 50% Black, 50% white; 1 is 100% white].^{50,51} Population density was calculated by dividing total block group population by block group area in square miles.

Statistical Analysis

Frequency distribution was generated for each variable (Table S1); given the high power of the sample size, only variables where the frequency across blocks was $\geq 10\%$ were considered for further analysis to prevent detecting significant covariates that have relatively minute effects.^{52,53} We used logistic regression to assess possible street block- and neighborhood-level correlates of the odds of an overdose event on a given block. Significant covariates in univariable analysis (at $p < 0.05$) were assessed in the multivariable model. To select the best fitting and most parsimonious model, we calculated Akaike's Information Criterion (AIC) and checked for multicollinearity using variance inflation factors (VIF). We examined the change in AIC with the addition of each potential covariate to avoid model overfit and identify the best-fitting, most parsimonious model.⁵⁴ We calculated spatial semi-variograms to detect residual spatial variation not accounted for by covariates.⁵⁵ We performed analyses for each city separately and combined to explore generalizable and localized overdose risk factors. All analyses were conducted using R 3.6.2.

RESULTS

People who overdosed in New York City (mean=47.35 years, sd=13.14) were of similar age to people who overdosed in Chicago (mean=47.11 years, sd=12.25, $p=0.526$) (Table 1). Distribution by sex was also similar, with 77% of overdoses among men in both cities ($p=0.332$). A larger proportion of New York City overdoses were non-Latino white (NYC: $n=983$, 34.4% vs. Chicago: $n=499$, 29.8%) or Latino (NYC: $n=975$, 34.0% vs. Chicago: $n=227$, 13.5%), while a larger proportion of Chicago overdoses were non-Latino Black (NYC: $n=777$, 27.1% vs. Chicago: $n=936$, 55.8%). While distribution of controls across boroughs/regions were comparable to case distributions, there were statistically significant demographic differences between cases and controls in both cities (Table 1). In both cities, controls were in census block groups with significantly less deprivation, significantly higher median household incomes, significantly higher proportions of non-Latino white residents, and significantly lower population densities ($p < 0.001$).

We found regional differences in frequency of block features across cities. Variables with frequency $< 10\%$ were not included in univariable models (New York City: alleys, abandoned buildings, vacant lots; Chicago: parks, bike lanes, civic buildings, row homes, building construction, graffiti, alcohol ads, loitering) (Table S1).

For New York City, we evaluated 31 block characteristics with sufficient frequency in univariable analysis (Table 2). Of these, three were not significant ($p>0.05$) and not included in multivariable analysis (security alarm signs, outdoor recreation outlets, youth on the street). In multivariable analysis, the odds of a fatal overdose were significantly increased on blocks with $\geq 50\%$ apartment buildings, multi-family homes, nonresidential structures (e.g., parking lots), and bus stops, and significantly decreased on blocks with single family homes and row homes, controlling for building count and other street- and neighborhood-level covariates. Several indicators of neighborhood physical order/disorder were also significantly associated with fatal overdose in multivariable analysis, including unmaintained properties, trash in the street, poor sidewalk maintenance, warning signs (e.g., “no trespassing”), traffic calming features (e.g., speed bumps), and street cleaning signs. Adults loitering was the only significant social indicator, controlling for other street- and neighborhood-level covariates.

For Chicago, we evaluated 24 block characteristics with sufficient frequency in univariable analysis (Table 2). Of these, two were not significant ($p>0.05$) and not included in multivariable analysis (street cleaning signs, sidewalk maintenance). In multivariable analysis, the odds of a fatal overdose were significantly increased on blocks with apartment buildings, restaurants, alleys, and bus stops, controlling for building count and other street- and neighborhood-level covariates. Several indicators of neighborhood physical disorder were also significantly associated with fatal overdose in multivariable analysis, including abandoned buildings, vacant lots, traffic calming features, warning signs, and landscaping, controlling for building count and other street- and neighborhood-level covariates. Adults on the street was the only significant social indicator, controlling for other street- and neighborhood-level covariates.

For both New York City and Chicago multivariable models, neighborhood deprivation index and segregation (ICE) showed multicollinearity ($VIF>3$) in multivariable analysis, with the best-fitting model containing neighborhood deprivation index. Residual semi-variograms indicated no unexplained residual spatial variation.

For both cities combined, we evaluated 20 block characteristics with sufficient frequency in both cities in univariable analysis (Table 3). Of these, one variable (unmaintained properties) was not significant ($p>0.05$) and not included in multivariable analysis. In multivariable analysis, the odds of a fatal overdose were significantly increased on blocks with apartment buildings, multifamily homes, nonresidential structures, restaurants, and bus stops, controlling for building count and other street- and neighborhood-level covariates. Several indicators of neighborhood physical disorder were also significantly associated with fatal overdose in multivariable analysis, including trash in the street, trash in other open spaces, poor sidewalk maintenance, warning signs, traffic calming features, and street cleaning signs, controlling for building count and other street- and neighborhood-level covariates. Adults on the street and adults sitting on stoops/porches were both significant, controlling for other street- and neighborhood-level covariates. Neighborhood deprivation index and segregation (ICE) showed multicollinearity ($VIF>3$) in multivariable analysis, with the best-fitting model containing neighborhood deprivation index. In the final

multivariable model, population density was not significant. Residual semi-variograms for this model indicated no unexplained residual spatial variation.

DISCUSSION

This study used a geographic case-control design to describe the physical and social environment on urban street blocks where fatal opioid overdoses occurred, comparing blocks where overdoses occurred to blocks where no overdoses occurred over the same time period in two major US cities. We operationalized aspects of neighborhood physical and social disorder by using NIfETy to measure built and social environment features on case and control blocks. We modeled the association between street block features and odds of fatal opioid overdose on a block, controlling for macro-level socioeconomic processes (e.g., neighborhood deprivation) to explore modifiable aspects of neighborhood design independent of sociodemographics. While a growing body of research suggests that macro-level physical and social characteristics significantly impact drug use and fatal overdose,^{6,8,16,22,29,56} to our knowledge, this is the first study to examine micro aspects of neighborhoods related to opioid overdose by comparing blocks where fatal opioid overdoses occurred to blocks with no fatal opioid overdoses in the same time period.

Consistent with previous U.S. studies of built environment risk factors for overdose,^{6–8,10,22} we identified several significant indicators of neighborhood physical disorder associated with increased odds of fatal opioid overdose common to both cities. For both cities, odds of overdose were significantly increased on blocks with apartment buildings, bus stops, street trash, traffic calming features, and warning signs (e.g., “no trespassing”), controlling for other block- and neighborhood-level covariates. These variables were also significant in the combined model, along with trash in other open spaces (e.g., alleys) and poor sidewalk maintenance. In the combined model, odds of overdose were significantly decreased on blocks with street cleaning signs, controlling for other block- and neighborhood-level covariates. These findings are consistent with studies of other cities, which found that neighborhood features may increase overdose risk in and of themselves by providing opportunities for drug selling and consumption. For example, research from Philadelphia found that transit hubs such as bus stops may provide access to drug markets for non-residents.⁵⁷ Our findings are further supported by studies of neighborhood disorder, which found significant positive associations between opioid overdose hotspots and 311 nuisance calls for litter, poor street maintenance, and traffic signal problems in Columbus, Ohio.¹⁶

Several significant variables described the neighborhood social environment. In New York City, odds of an overdose were 41% higher on blocks with loitering adults; in Chicago, odds of an overdose were twice as high on blocks with adults on the street. In the combined model, odds of an overdose were 37% higher on blocks with adults on the street and 34% higher on blocks with adults sitting on stoops/porches, controlling for other street- and neighborhood-level variables. To our knowledge, few studies have disaggregated the street social environment from neighborhood sociodemographics or other macro-level social forces such as social networks which influence norms around drug use.^{7,56,58,59} Several studies have shown that long-term exposure to neighborhood social disorder increases

chronic stress, which may in turn influence drug use.⁶⁰ Further investigation into how and why the physical manifestations of social environments impact fatal overdose is warranted.

Several block characteristics which showed low frequency in one city were significantly positively associated with odds of a fatal overdose on a block in the other city. For example, in Chicago the odds of an overdose were 50% higher on blocks with abandoned buildings and 2.8 times higher on blocks with vacant lots, controlling for other street- and neighborhood-level covariates. Findings are consistent with previous studies identifying abandoned or dilapidated housing^{6,8} and vacant lots²⁹ as significantly positively associated with fatal overdose as vacant lots and abandoned buildings may be locations for open-air drug markets, particularly in more resource-deprived neighborhoods.²⁹ However, these street micro-characteristics were rare in New York City, where only 5.6% of blocks had abandoned buildings and 6.5% of blocks had vacant lots (Chicago: 17.8% and 21.3%, respectively). Furthermore, several block characteristics were significantly associated with odds of fatal overdose in one city and not significant in the other. For example, restaurants were not significantly associated with odds of overdose in multivariable analysis in New York City but were almost three times more likely on overdose blocks in Chicago, controlling for other street- and neighborhood-level covariates. While we could find no peer reviewed studies on overdoses and restaurants specifically, news reports suggest that overdoses in restaurants are increasing.⁶¹ Furthermore, there is evidence of a positive association between density of alcohol outlets (including restaurants, bars, and other nightlife venues) and neighborhood overdose rate.²² Zoning ordinances may also impact differences between cities; for example, restaurants and other alcohol-serving establishments may be allowed to be close to certain features that predict or explain overdose in one city but not the other.^{62,63} These regional differences reinforce the importance of tailoring research questions and interventions to specific people, places, and circumstances, particularly when investigating physical and social environment risk factors for overdose.

This case-control study was limited to fatal opioid overdose locations in two major US cities. Neighborhood risk factors for opioid overdose may differ across cities and by urbanicity. As we did not have access to nonfatal overdose data, we could not measure street block characteristics at locations of nonfatal overdoses or explore neighborhood factors that might reduce overdose fatality. Only New York City OCME records included “found location type,” which provided a brief description of where the overdose occurred; we could not investigate the type of location where an overdose occurred or if an overdose occurred in a public place outside a person who use drug’s home or a friend or family member’s residence (e.g., on the street, in a park).⁶⁴ Previous New York City studies over the same time period found that 11% of fatal opioid overdoses occurred in public places.⁵⁶ GSV images do not capture fine details such as drug paraphernalia on the street or time-varying measures such noise, which may limit our ability to describe the street block social environment or drug use environment.⁴⁷ It is possible that temporal delays in GSV image capture compared to overdose occurrence may affect reliability.⁶⁵ GSV images were available within six months for over three-quarters of overdose locations, a high rate of coverage compared to other studies using GSV.⁴⁷ Neighborhood characteristics change slowly over several years, suggesting strong reliability for retrospective street block measures.^{66,67}

CONCLUSION

This study identified clear patterns in street block-level characteristics associated with fatal opioid overdose in two major US cities. Our study supports findings from previous studies of overdose in urban areas about the unique contribution of neighborhood social and physical characteristics as important risk factors for fatal opioid overdose over and above sociodemographics.^{6–9,11,16,56} Our study also provides further evidence to support community infrastructure reinvestment programs targeting vacant lot and abandoned building remediation to improve community health and well-being.^{28–30} Further inquiry is needed to understand the mechanisms by which street micro-characteristics impact opioid overdose and the context in which these mechanisms operate. Better understanding of how and why neighborhood features impact fatal opioid overdose may identify novel community development strategies for overdose prevention.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability Statement:

New York City fatal overdose data are the property of the New York City Office of the Chief Medical Examiner and are available from them upon request. Cook County Medical Examiner data are publicly available for download at <https://datacatalog.cookcountyil.gov/d/cjeq-bs86>

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Table 1.

Description of sociodemographic characteristics of cases at the time of fatal opioid overdose and control blocks where no fatal opioid overdoses occurred in the same time period, New York City and Chicago, 2018–2019

Variable	New York City			Chicago			p *
	Cases (n=2867)	Controls (n=2093)	p *	Cases (n=1677)	Controls (n=1148)	p *	
Individual characteristics							
Age (mean ± SD)	47.35 ± 13.14	--	--	47.11 ± 12.25	--	--	0.526
Gender, n (%)		--	--		--	--	0.332
Male	2193 (76.5)			1305 (77.8)			
Female	667 (23.3)			371 (15.9)			
Trans	7 (0.2)			1 (0.0)			
Race/ethnicity, n (%)		--	--		--	--	<0.001
Non-Latino white	983 (34.4)			499 (29.8)			
Non-Latino Black	777 (27.1)			936 (55.8)			
Latino	975 (34.0)			227 (13.5)			
Other	132 (4.6)			15 (0.89)			
Block characteristics							
Borough/region, n (%) [‡]			--			--	--
1	697 (24.3)	517 (24.7)		115 (6.9)	77 (6.7)		
2	870 (30.4)	639 (30.5)		669 (39.9)	459 (40.0)		
3	653 (22.8)	476 (22.7)		563 (33.6)	386 (33.6)		
4	452 (15.8)	324 (15.5)		330 (19.7)	226 (19.7)		
5 (NYC only)	195 (6.8)	137 (6.5)					
Count of structures on block, mean ± SD	16.72±14.60	13.74±13.65	<0.001	19.20±13.45	11.62±12.41	<0.001	--
Neighborhood deprivation index (range: −5 to +5), mean ± SD	0.50±1.67	−0.52±1.52	<0.001	0.51±1.65	−1.02±1.40	<0.001	--
Segregation (ICE) (range: −1 to +1), mean ± SD	0.06±0.50	0.22±0.56	<0.001	−0.29±0.71	0.28±0.65	<0.001	--
Median household income (in \$10,000s), mean ± SD	5.67±3.57	7.57±4.01	<0.001	4.64±3.03	7.00±3.72	<0.001	--
Population density per square mile (in 1,000s), mean ± SD	70.33±54.68	49.83±46.66	<0.001	18.06±16.52	13.88±11.36	<0.001	--

* Pearson chi-square test of independence for categorical variables or Welch two sample t-test for continues variables

[‡] New York City borough codes: 1=Manhattan, 2=Bronx, 3=Brooklyn, 4=Queens, 5=Staten Island. Chicago region codes: 1=Downtown: Chicago Central and Loop, 2=Chicago South Side, 3=Chicago West Side, 4=Chicago North Side

Univariable and multivariable logistic regression results for odds of fatal opioid overdose occurring on a given block, New York City and Chicago, 2018–2019

Table 2.

Variable	NYC unadjusted OR (95%CI)	p	NYC adjusted aOR (95%CI) *	p	Chicago unadjusted OR (95%CI)	p	Chicago adjusted aOR (95%CI) *	p
Types of structures								
Apartment buildings (Ref: 0%)								
<50% of block	2.14 (1.78, 2.58)	<0.001	1.16 (0.90, 1.50)	0.250	4.78 (3.82, 6.01)	<0.001	2.01 (1.50, 2.74)	<0.001
≥50% of block	3.32 (2.92, 3.79)	<0.001	2.00 (1.58, 2.52)	<0.001	4.24 (3.53, 5.11)	<0.001	2.77 (2.14, 3.60)	<0.001
Multi-family homes (Ref: 0%)							---	
<50% of block	1.36 (1.17, 1.59)	<0.001	1.47 (1.19, 1.82)	<0.001	2.23 (1.81, 2.75)	<0.001		
≥50% of block	1.56 (1.24, 1.97)	<0.001	1.80 (1.40, 2.58)	<0.001	1.37 (1.06, 1.79)	0.0184		
Single family homes (Ref: 0%)							---	
<50% of block	0.84 (0.72, 0.98)	0.030	0.94 (0.74, 1.18)	0.564	1.71 (1.36, 2.18)	<0.001		
≥50% of block	0.38 (0.33, 0.45)	<0.001	0.63 (0.49, 0.82)	0.001	0.56 (0.47, 0.66)	<0.001		
Row homes (Ref: 0%)								
<50% of block	0.74 (0.62, 0.88)	<0.001	0.53 (0.42, 0.68)	<0.001	Low frequency			
≥50% of block	0.37 (0.31, 0.44)	<0.001	0.31 (0.24, 0.41)	<0.001				
Commercial structures (Ref: 0%)			---				---	
<50% of block	2.18 (1.91, 2.49)	<0.001			2.80 (2.30, 3.43)	<0.001		
≥50% of block	1.50 (1.29, 1.75)	<0.001			1.67 (1.37, 2.03)	<0.001		
Nonresidential structures (Ref: No)							---	
Restaurants (Ref: No)	2.02 (1.77, 2.32)	<0.001	1.58 (1.33, 1.88)	<0.001	1.63 (1.38, 1.93)	<0.001		
Alleys (Ref: No)	1.61 (1.42, 1.84)	<0.001	---		3.59 (2.73, 4.78)	<0.001	2.92 (2.05, 4.20)	<0.001
Bus stops (Ref: No)	Low frequency				1.49 (1.32, 1.69)	<0.001	1.41 (1.13, 1.77)	0.003
Bike lane(s) (Ref: No)	1.70 (1.47, 1.97)	<0.001	1.70 (1.41, 2.06)	<0.001	4.78 (3.69, 6.21)	<0.001	4.44 (3.18, 6.27)	<0.001
Parks (Ref: No)	1.18 (1.00, 1.39)	0.046			Low frequency			
Places of worship (Ref: No)	1.33 (1.10, 1.60)	0.003			Low frequency			
Civic buildings (Ref: No)	1.39 (1.18, 1.65)	<0.001			3.04 (2.29, 4.09)	<0.001	---	
Physical order and disorder					Low frequency			

Variable	NYC unadjusted OR (95%CI)	p	NYC adjusted aOR (95%CI)*	p	Chicago unadjusted OR (95%CI)	p	Chicago adjusted aOR (95%CI)*	p
Abandoned buildings (Ref: No)	Low frequency				7.58 (5.69, 10.29)	<0.001	1.51 (1.04, 2.22)	0.031
Vacant lots (Ref: No)	Low frequency				9.81 (7.39, 13.27)	<0.001	2.82 (1.96, 4.13)	<0.001
Unmaintained properties (Ref: No)	0.86 (0.75, 0.97)	0.018	0.38 (0.32, 0.46)	<0.001	1.69 (1.43, 2.01)	<0.001	---	---
Trash in street (Ref: No)	3.02 (2.66, 3.43)	<0.001	2.29 (1.95, 2.70)	<0.001	3.73 (3.06, 4.56)	<0.001	1.41 (1.08, 1.86)	0.013
Trash in other open spaces (Ref: No)	1.91 (1.66, 2.19)	<0.001	---		4.40 (3.58, 5.44)	<0.001	---	---
Poor sidewalk maintenance (Ref: No)	1.70 (1.45, 2.01)	<0.001	1.82 (1.46, 2.26)	<0.001	1.20 (0.97, 1.49)	0.093	---	---
Warning signs (Ref: No)	2.57 (2.28, 2.89)	<0.001	1.90 (1.63, 2.21)	<0.001	2.81 (2.37, 3.34)	<0.001	1.35 (1.07, 1.72)	0.013
Security alarm signs (Ref: No)	0.90 (0.79, 1.02)	0.106			1.34 (1.15, 1.57)	<0.001	---	---
Traffic calming features (Ref: No)	1.92 (1.65, 2.23)	<0.001	1.28 (1.06, 1.54)	0.011	2.39 (2.00, 2.88)	<0.001	1.41 (1.11, 1.80)	0.006
Landscaping (Ref: No)	0.74 (0.66, 0.83)	<0.001			0.51 (0.43, 0.61)	<0.001	0.53 (0.41, 0.69)	<0.001
Street cleaning signs (Ref: No)	0.76 (0.68, 0.85)	<0.001	0.36 (0.31, 0.42)	<0.001	1.21 (0.99, 1.49)	0.057	---	---
Building construction (Ref: No)	1.23 (1.08, 1.40)	0.002	---		Low frequency			
Graffiti (Ref: No)	1.27 (1.12, 1.43)	<0.001	---		Low frequency			
Alcohol ads (Ref: No)	2.28 (1.86, 2.82)	<0.001	---		Low frequency			
Social order and disorder								
Adults on the street (Ref: No)	2.13 (1.82, 2.50)	<0.001	--		4.02 (3.43, 4.72)	<0.001	2.04 (1.65, 2.53)	<0.001
Adults loitering (Ref: No)	2.52 (2.10, 3.04)	<0.001	1.49 (1.19, 1.87)	<0.001	Low frequency			
Adults sitting on stoop (Ref: No)	2.01 (1.70, 2.40)	<0.001	--		4.24 (3.22, 5.65)	<0.001	---	---
Youth on the street (Ref: No)	1.06 (0.93, 1.20)	0.375			2.18 (1.73, 2.77)	<0.001	---	---
Neighborhood-level								
Median Household Income (in \$10,000s)	0.88 (0.86, 0.89)	<0.001	0.95 (0.92, 0.97)	<0.001	0.81 (0.79, 0.83)	<0.001	0.99 (0.94, 1.03)	0.526
Neighborhood Deprivation Index (range: -5 to +5) [‡]	1.48 (1.43, 1.54)	<0.001	1.23 (1.15, 1.32)	<0.001	1.86 (1.76, 1.97)	<0.001	1.56 (1.41, 1.72)	<0.001
Segregation (ICE) (range: -1 to +1)	0.56 (0.50, 0.63)	<0.001	---		0.32 (0.29, 0.36)	<0.001	---	---
Population density (1000s per sq mile)	1.01 (1.01, 1.01)	<0.001	1.00 (1.00, 1.00)	0.008	1.03 (1.02, 1.04)	<0.001	1.02 (1.01, 1.02)	<0.001
Building count (n)	1.02 (1.01, 1.02)	<0.001	1.04 (1.03, 1.04)	<0.001	1.05 (1.04, 1.05)	<0.001	1.06 (1.05, 1.07)	<0.001

[‡] Odds ratio denotes a 1.00-unit change in the score over the -5 to +5 scale

^{*} Adjusted for other covariates in the column

Low frequency: Only variables with ≥10% frequency were included in analyses. See Supplemental Table S1 for item definitions and distribution.

Table 3.

Univariable and multivariable logistic regression results for odds of fatal opioid overdose occurring on a given block for both New York City and Chicago combined, 2018–2019

Variable	Combined, unadjusted OR (95% CI)	p	Combined, adjusted aOR (95% CI) *	p *
Types of structures				
Apartment buildings (Ref: 0%)				
<50% of block	2.97 (2.58, 3.42)	<0.001	1.39 (1.16, 1.67)	<0.001
≥50% of block	3.42 (3.09, 3.79)	<0.001	2.68 (2.30, 3.11)	<0.001
Multi-family homes (Ref: 0%)				
<50% of block	1.64 (1.45, 1.85)	<0.001	1.23 (1.04, 1.44)	0.013
≥50% of block	1.47 (1.24, 1.75)	<0.001	1.58 (1.27, 1.96)	<0.001
Single family homes (Ref: 0%)				
<50% of block	1.10 (0.97, 1.26)	0.130	---	
≥50% of block	0.50 (0.45, 0.56)	<0.001		
Commercial structures (Ref: 0%)			---	
<50% of block	2.18 (1.96, 2.42)	<0.001		
≥50% of block	1.51 (1.34, 1.70)	<0.001		
Nonresidential structures (Ref: No)	1.86 (1.68, 2.07)	<0.001	1.54 (1.35, 1.76)	<0.001
Restaurants (Ref: No)	1.84 (1.64, 2.07)	<0.001	1.43 (1.23, 1.67)	<0.001
Bus stops (Ref: No)	2.29 (2.02, 2.60)	<0.001	2.58 (2.20, 3.03)	<0.001
Places of worship (Ref: No)	1.72 (1.49, 1.98)	<0.001		
Physical order and disorder				
Unmaintained properties (Ref: No)	1.10 (0.99, 1.22)	0.061	---	
Trash in street (Ref: No)	3.15 (2.83, 3.50)	<0.001	1.73 (1.51, 1.99)	<0.001
Trash in other open spaces (Ref: No)	2.53 (2.26, 2.85)	<0.001	1.19 (1.02, 1.39)	0.027
Poor sidewalk maintenance (Ref: No)	1.49 (1.31, 1.70)	<0.001	1.18 (1.00, 1.39)	0.050
Warning signs (Ref: No)	2.58 (2.35, 2.85)	<0.001	1.58 (1.40, 1.78)	<0.001
Security alarm signs (Ref: No)	1.07 (0.97, 1.18)	0.168		
Traffic calming features (Ref: No)	2.10 (1.88, 2.37)	<0.001	1.50 (1.30, 1.73)	0.011
Landscaping (Ref: No)	0.68 (0.62, 0.75)	<0.001		
Street cleaning signs (Ref: No)	0.85 (0.77, 0.93)	<0.001	0.37 (0.32, 0.42)	<0.001
Social order and disorder				
Adults on the street (Ref: No)	2.61 (2.35, 2.91)	<0.001	1.37 (1.18, 1.58)	0.001
Adults sitting on stoop (Ref: No)	2.53 (2.19, 2.93)	<0.001	1.34 (1.12, 1.60)	0.013
Youth on the street (Ref: No)	1.24 (1.12, 1.39)	<0.001		
Neighborhood-level				
Median Household Income (in \$10,000s)	0.86 (0.85, 0.87)	<0.001	0.96 (0.94, 0.98)	<0.001
Neighborhood Deprivation Index (range: -5 to +5) [‡]	1.59 (1.54, 1.64)	<0.001	1.35 (1.28, 1.43)	<0.001
Segregation (ICE) (range: -1 to +1)	0.43 (0.40, 0.46)	<0.001	---	---

Variable	Combined, unadjusted OR (95% CI)	p	Combined, adjusted aOR (95% CI) *	p *
Population density (1000s per sq mile)	1.01 (1.01, 1.01)	<0.001	1.00 (0.99, 1.00)	0.483
Building count (n)	1.03 (1.02, 1.03)	<0.001	1.03 (1.02, 1.03)	<0.001

[#] Odds ratio denotes a 1.00-unit change in the score over the -5 to +5 scale

* Adjusted for other covariates in the column.

Note: Only variables with ≥10% frequency in both cities were included in analyses. AIC: 7428.8